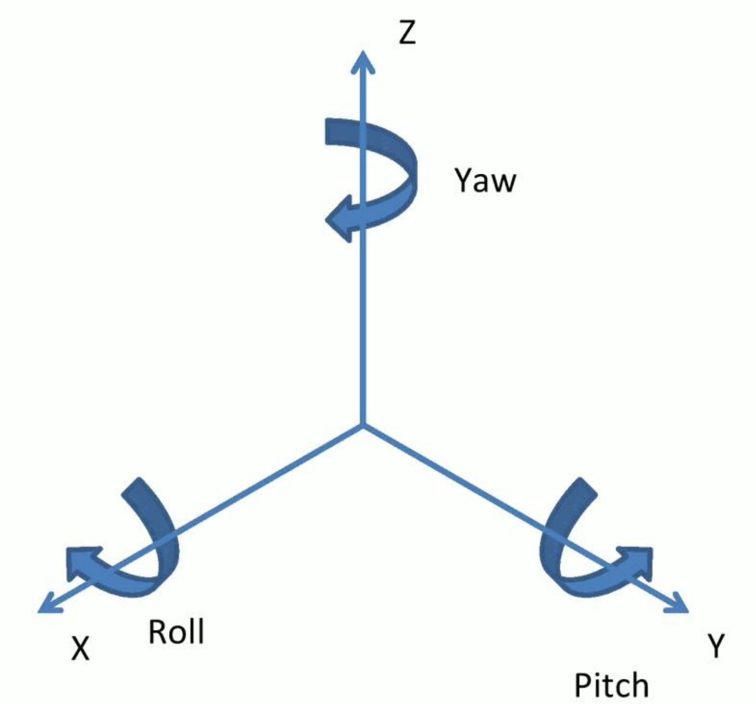
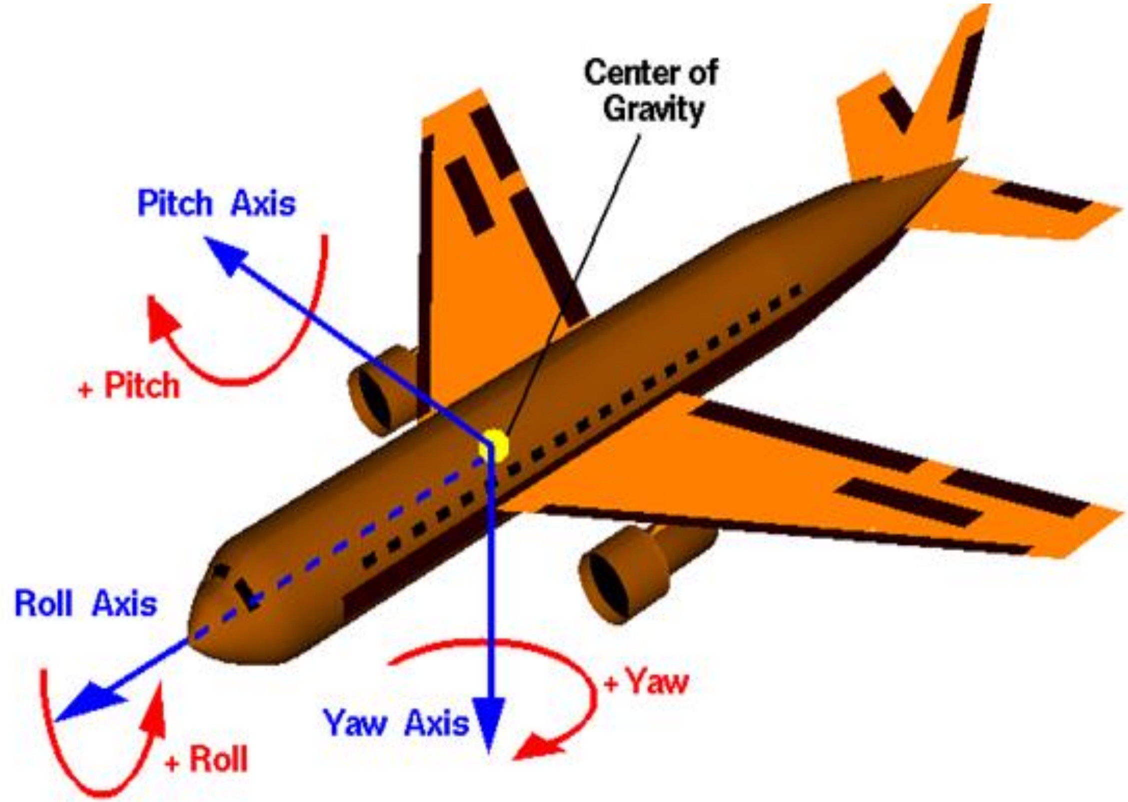
ROTATIONS

Introduction

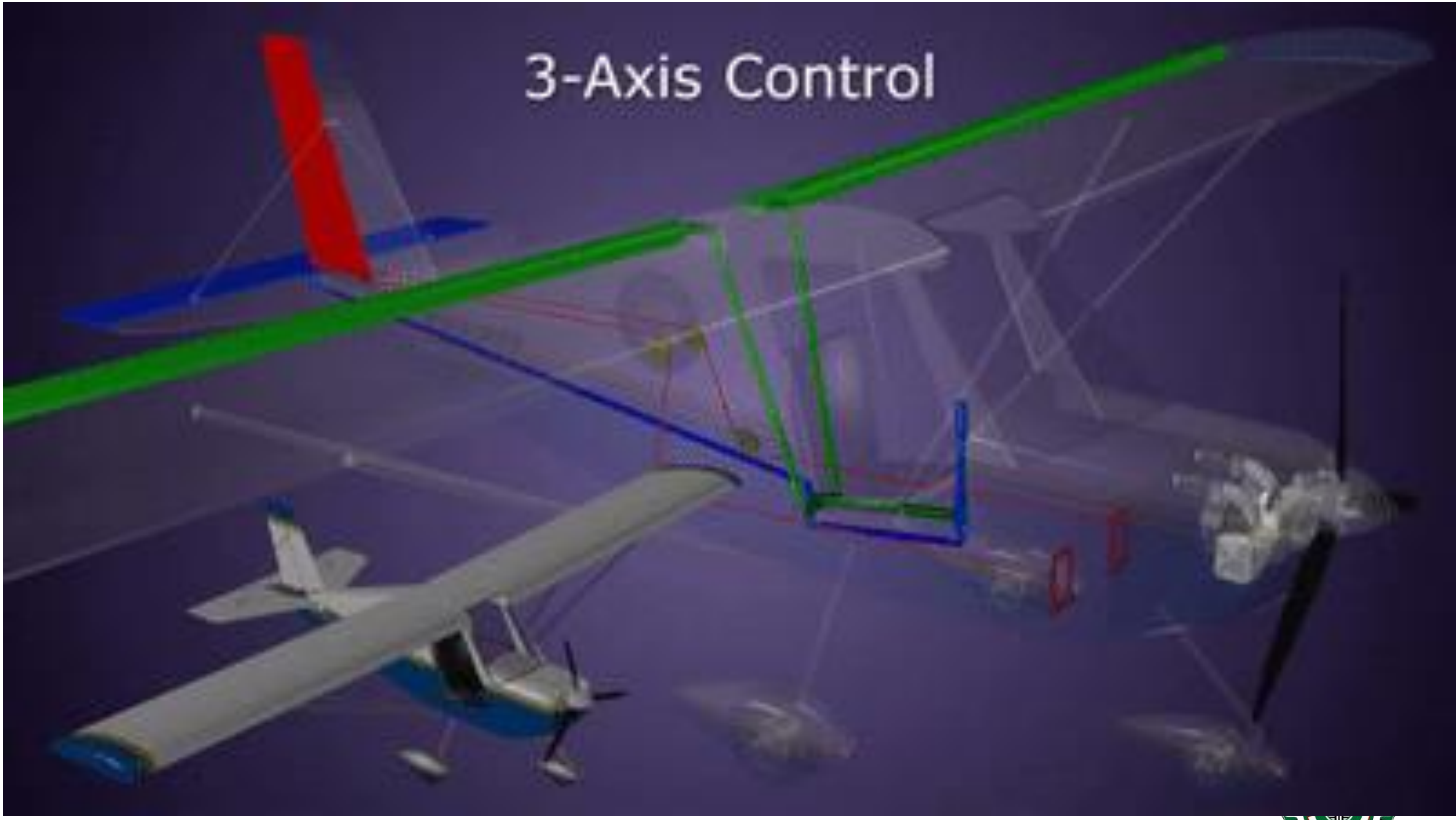
- The **XYZ** rotation sequence is called ***roll - pitch - yaw***, or just ***RPY***.
- The sequence is intuitive when describing the attitude of moving machines such as ships, aircraft, cars, and robots.
- The X-axis is considered to be the robot's forward direction.
- The Z-axis is usually oriented in the upward direction; however, with aircraft the Z-axis is often in the downward direction.
- Rotation about the X-axis is called *roll*.
- Rotation about the Y-axis is called *pitch*.
- Rotation about the Z-axis is called *yaw*.

Roll, Pitch and Yaw (RPY) rotations

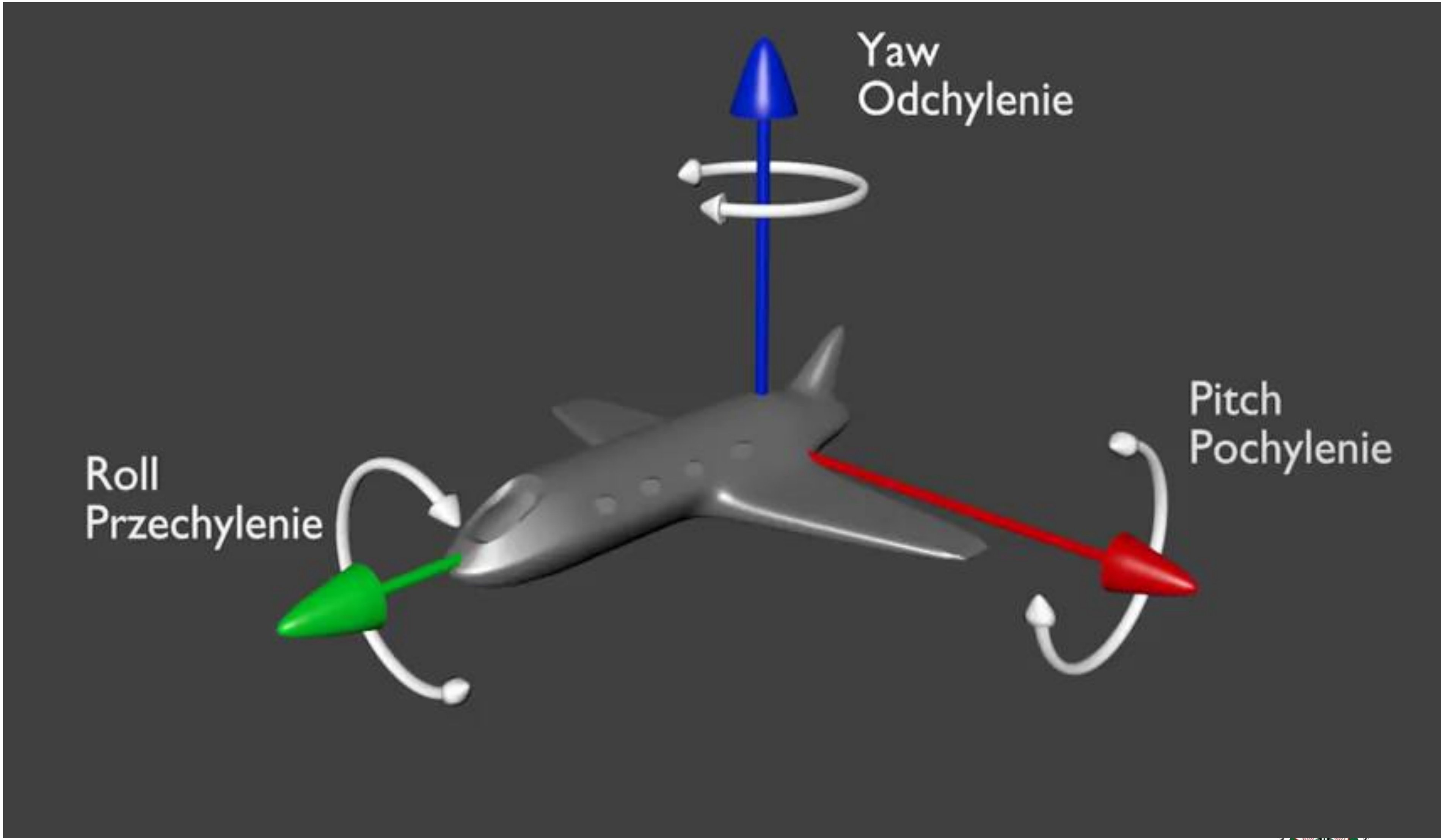




Roll, Pitch and Yaw (RPY) rotations



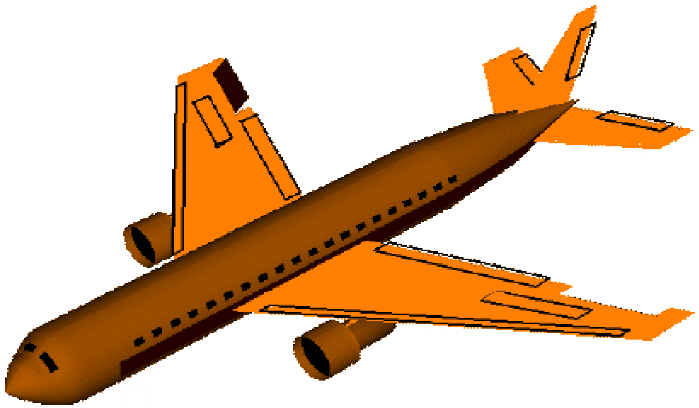
Roll, Pitch and Yaw (RPY) rotations

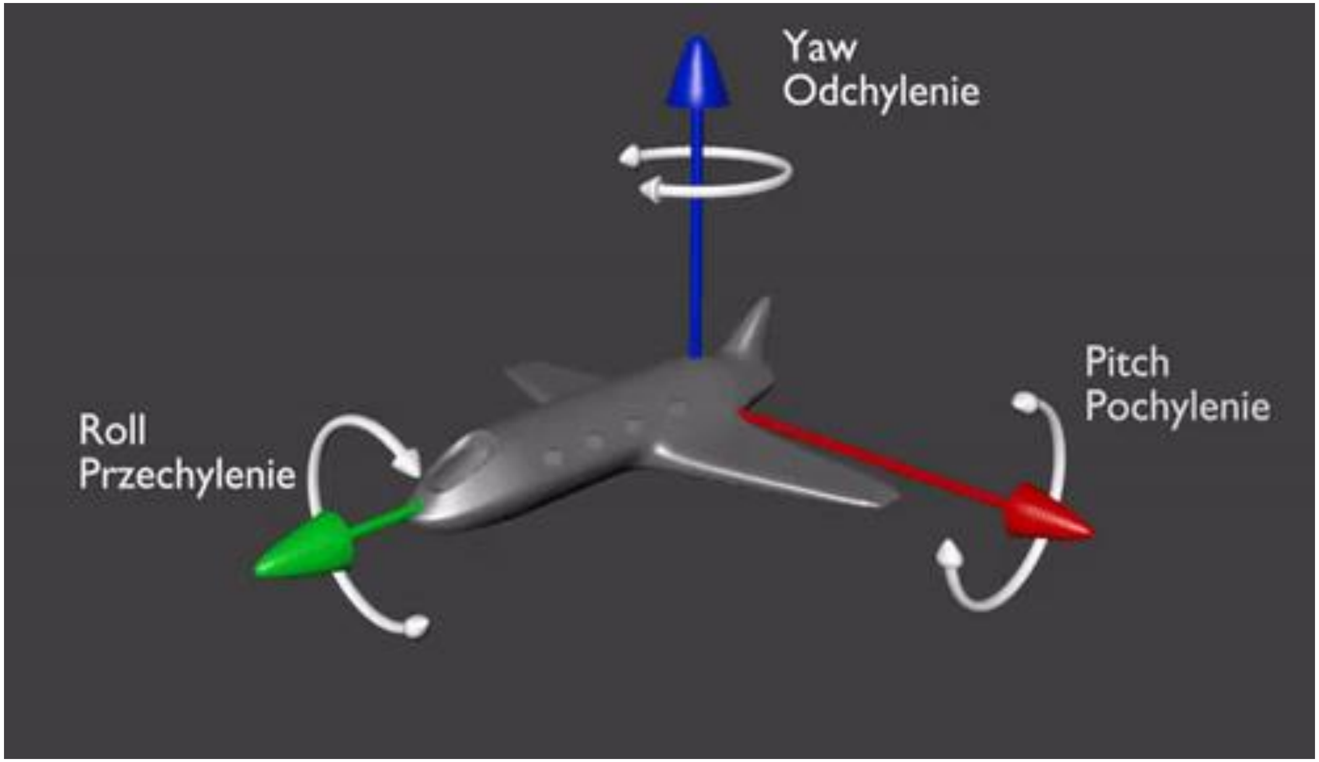


ROLL (Rotation about the X-Axis)

- A roll is a counterclockwise rotation of θ about the x-axis.
- A roll moment is a force that attempts to cause a system to rotate about its X axis, from side-to-side.
- A good example of roll is an airplane banking (This maneuver is used to change the *aircraft* heading.)
- The rotation matrix is given by

$$R_x(\theta_x) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta_x & -\sin\theta_x \\ 0 & \sin\theta_x & \cos\theta_x \end{bmatrix}$$




Yaw
Odchylenie
Pitch
Pochylenie
Roll
Przechylenie

PITCH (Rotation about the Y-Axis)

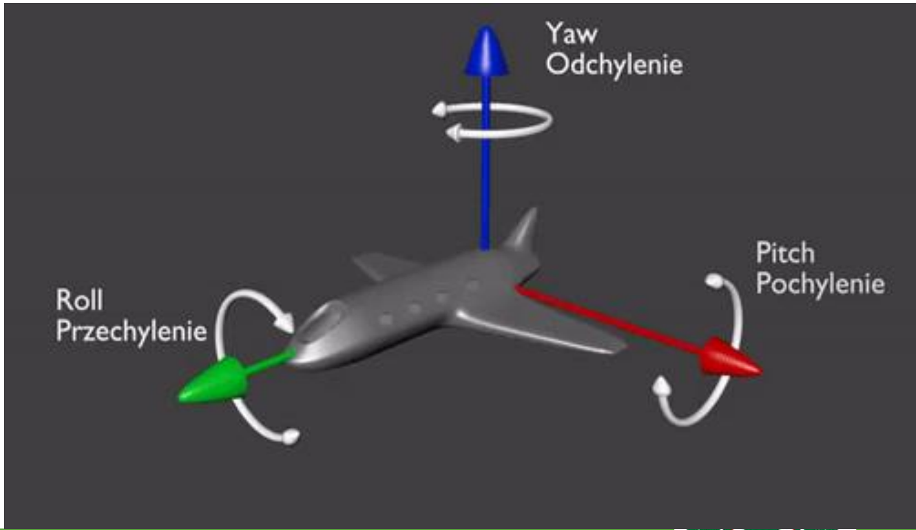
- A pitch is a counterclockwise rotation of θ about the y-axis.
- A pitch moment attempts to cause a system to rotate about its Y axis, from front to back.
- To envision pitch, think of the nose of an airplane pointing downward or upward.
- The rotation matrix is given by

$$R_y(\theta_y) = \begin{bmatrix} \cos\theta_y & 0 & \sin\theta_y \\ 0 & 1 & 0 \\ -\sin\theta_y & 0 & \cos\theta_y \end{bmatrix}$$

YAW (Rotation about the Z-Axis)

- Rotation around the vertical axis is called yaw
- A yaw is a counterclockwise rotation of θ about the $\boldsymbol{z-axis}$.
- Yaw occurs when a force attempts to cause a system to rotate about its Z axis.
- To visualize yaw, imagine a model airplane suspended on a string. If the wind blows just right, the airplane's wings and nose will remain level (no rolling or pitching), but it will rotate around the string from which it's suspended. This is yaw.
- The rotation matrix is given by:

$$R_z(\theta_z) = \begin{bmatrix} \cos\theta_z & -\sin\theta_z & 0 \\ \sin\theta_z & \cos\theta_z & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



2D Rotation

- Note that the upper left entries of ***Rz(θz)*** form a 2D rotation applied to the x and y coordinates, whereas the coordinate remains constant.

$$R_z(\theta_z) = \begin{bmatrix} \boxed{\begin{matrix} \cos\theta_z & -\sin\theta_z & 0 \\ \sin\theta_z & \cos\theta_z & 0 \end{matrix}} \\ \begin{matrix} 0 & \quad 0 \quad & 1 \end{matrix} \end{bmatrix}$$

Rotation Matrix Computation

Eg. Compute the rotation matrix of a mobile robot that rotates at an angle of 30 degrees about the X-Axis, Y-Axis and Z- Axis

Rotation about the X-Axis Computation (Roll)

Angle=θ=30 degrees

$$R_x(\theta_x) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta_x & -\sin\theta_x \\ 0 & \sin\theta_x & \cos\theta_x \end{bmatrix}$$

$$R_x(30°) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos 30° & -\sin 30° \\ 0 & \sin 30° & \cos 30° \end{bmatrix}$$

$$R_x(30°) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0.8660 & -0.5000 \\ 0 & 0.5000 & 0.8660 \end{bmatrix}$$

Rotation about the Y-Axis Computation (Pitch)

Angle=θ=30 degrees

$$R_y(\theta_y) = \begin{bmatrix} \cos\theta_y & 0 & \sin\theta_y \\ 0 & 1 & 0 \\ -\sin\theta_y & 0 & \cos\theta_y \end{bmatrix}$$

$$R_y(30°) = \begin{bmatrix} \cos 30° & 0 & \sin 30° \\ 0 & 1 & 0 \\ -\sin 30° & 0 & \cos 30° \end{bmatrix}$$

$$R_y(30°) = \begin{bmatrix} 0.8660 & 0 & 0.5000 \\ 0 & 1 & 0 \\ -0.5000 & 0 & 0.8660 \end{bmatrix}$$

Rotation about the Y-Axis Computation (Yaw)

$$R_z(\theta_z) = \begin{bmatrix} \cos\theta_y & -\sin\theta_y & 0 \\ \sin\theta_y & \cos\theta_y & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$R_z(30°) = \begin{bmatrix} \cos 30\,° & -\sin 30° & 0 \\ \sin 30° & \cos 30° & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$R_z(30°) = \begin{bmatrix} 0.8660 & -0.5000 & 0 \\ 0.5000 & 0.8660 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

END OF LECTURE

Wait for an assignment

ASSIGNMENT 3

1. Using appropriate matrices, explain the following:
    i. Roll
    ii. Pitch
    iii. Yaw
2. Compute the rotation matrix of an aerial autonomous vehicle that rotates at an angle of 70 degrees about the:
    i. X-axis
    ii. Y-axis
    iii. Z-axis